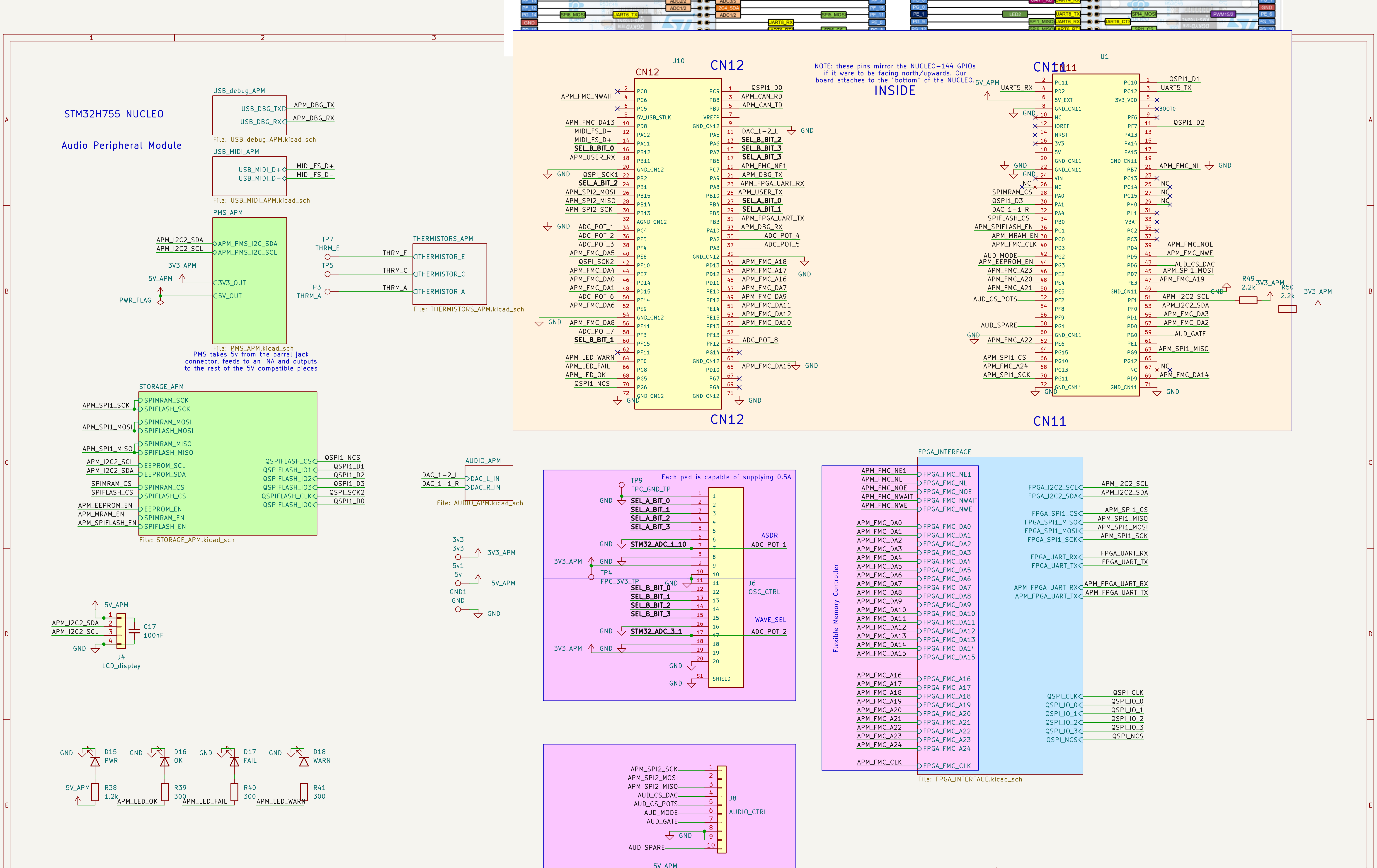
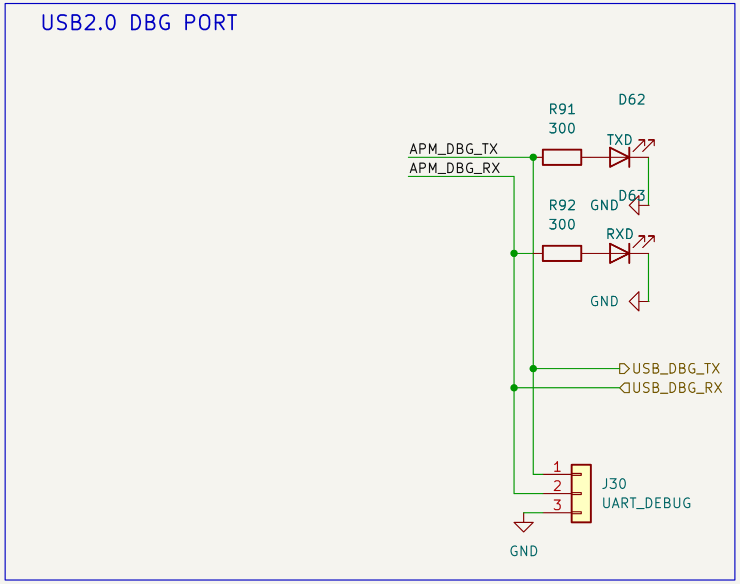


Sheet: /
File: APM.kicad_sch
Title:
Size: A3 Date: Rev:
KiCad E.D.A. 10.0.4 Id: 1/8

UART -> USB Debug Port

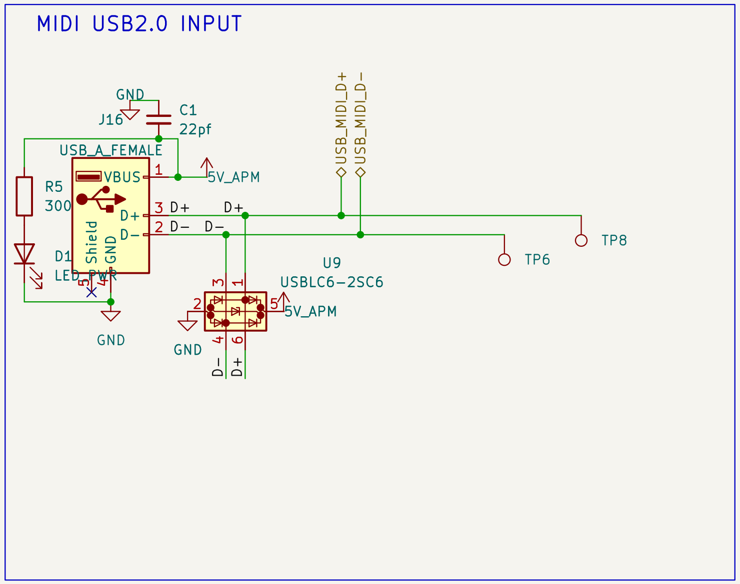
UART debug messages avaiable via a
USB port and a 3 pin jumper for
TX, RX, and GND UART connections



Sheet: /USB_debug_APM/
File: USB_debug_APM.kicad_sch

Title:

Size: A4	Date:	**Rev:**
KiCad E.D.A. 10.0.4		Id: 2/8



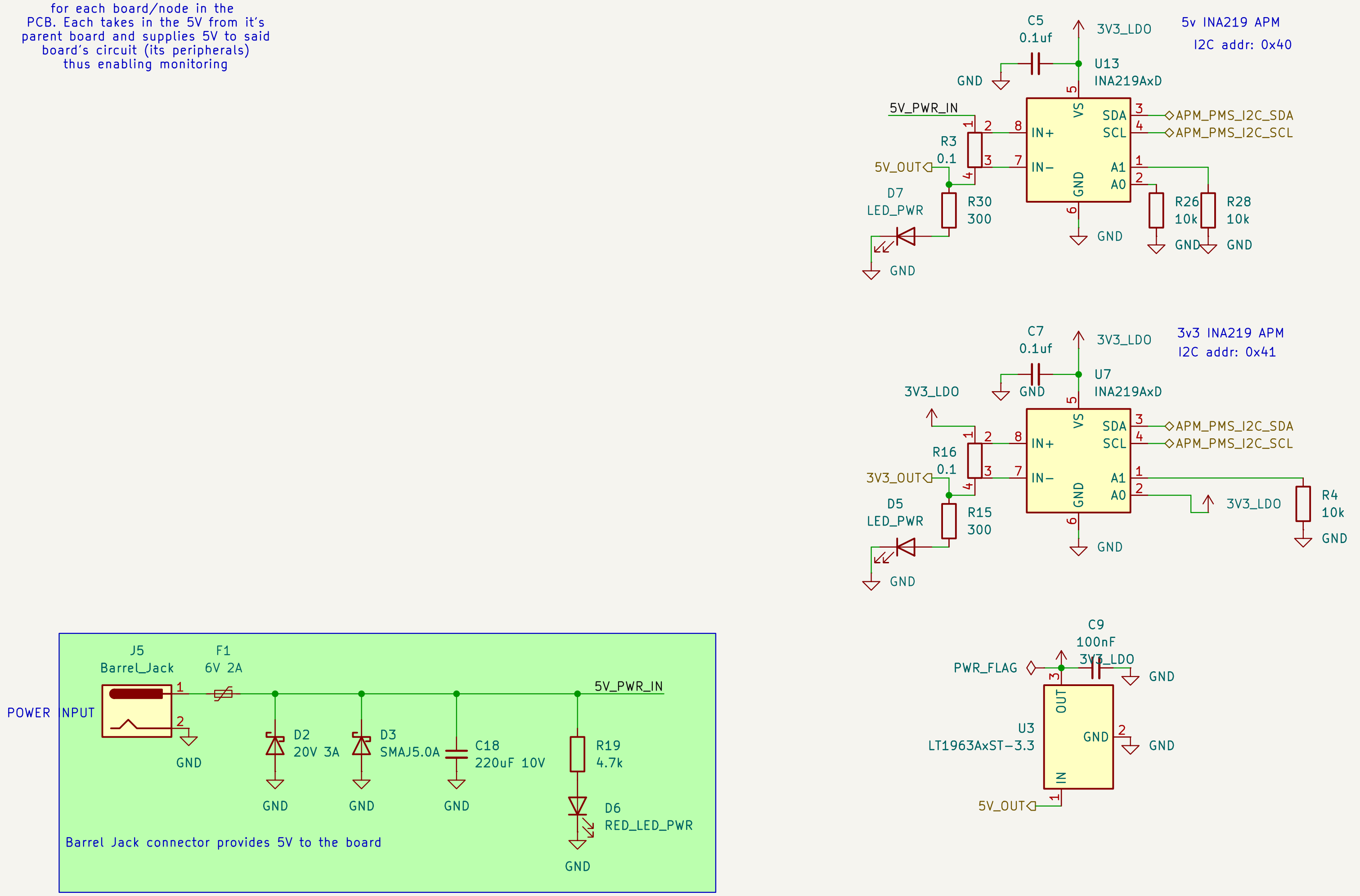
Sheet: /USB_MIDI_APM/
File: USB_MIDI_APM.kicad_sch

Title:

Size: A4	Date:	Rev:
KiCad E.D.A. 10.0.4		Id: 3/8

Power Monitor and Supply

PMS has an INA219 current sensor for each board/node in the PCB. Each takes in the 5V from it's parent board and supplies 5V to said board's circuit (its peripherals) thus enabling monitoring

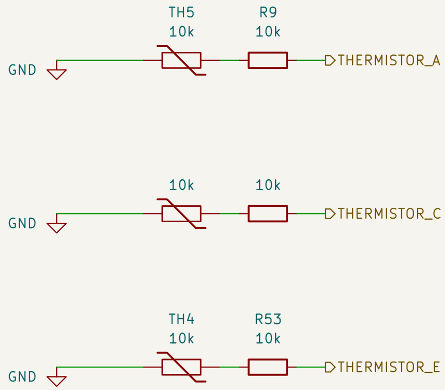


Sheet: /PMS_APM/
File: PMS_APM.kicad_sch

Title:

Size: A4 Date: **Rev:**

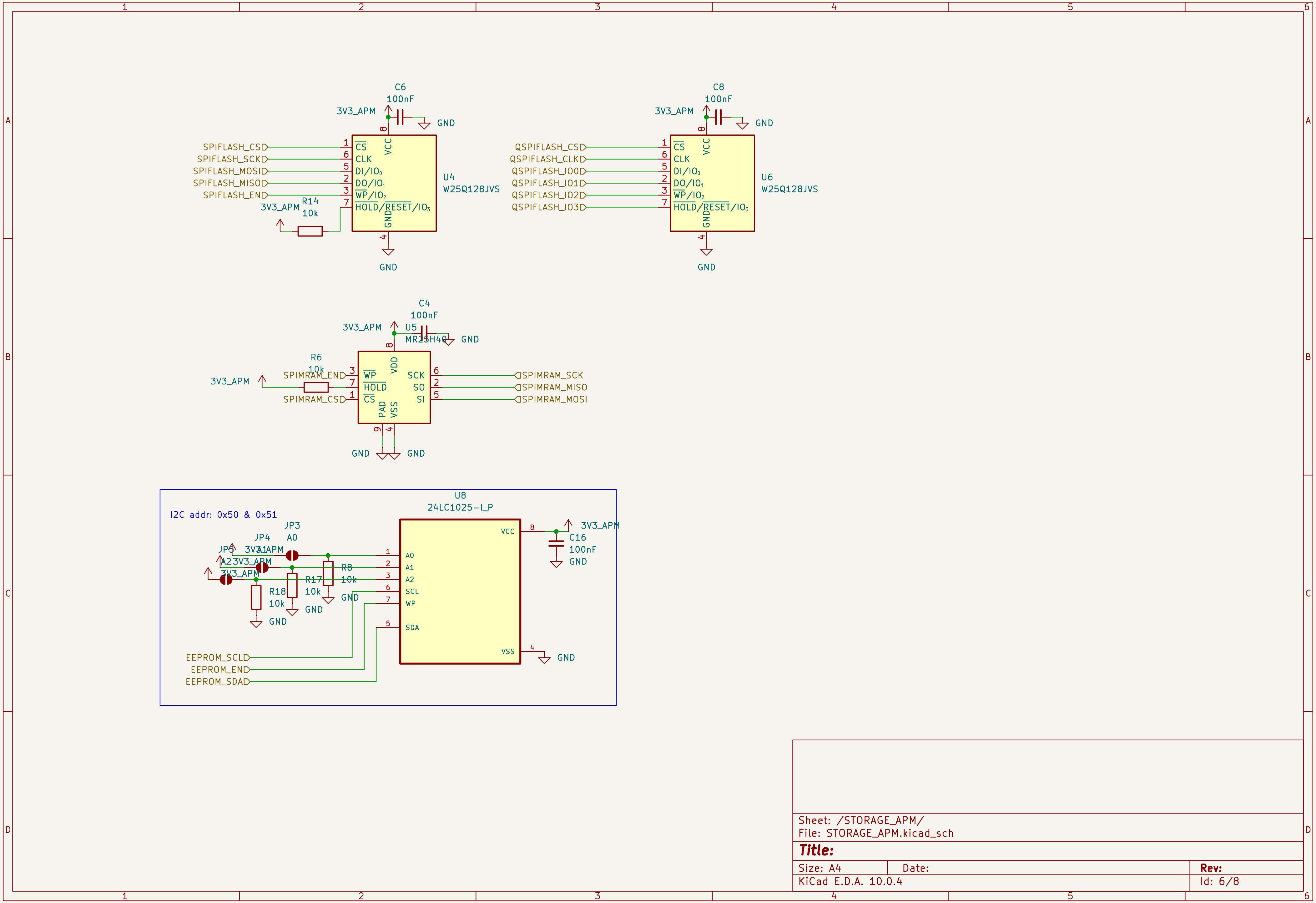
KiCad E.D.A. 10.0.4 Id: 4/8

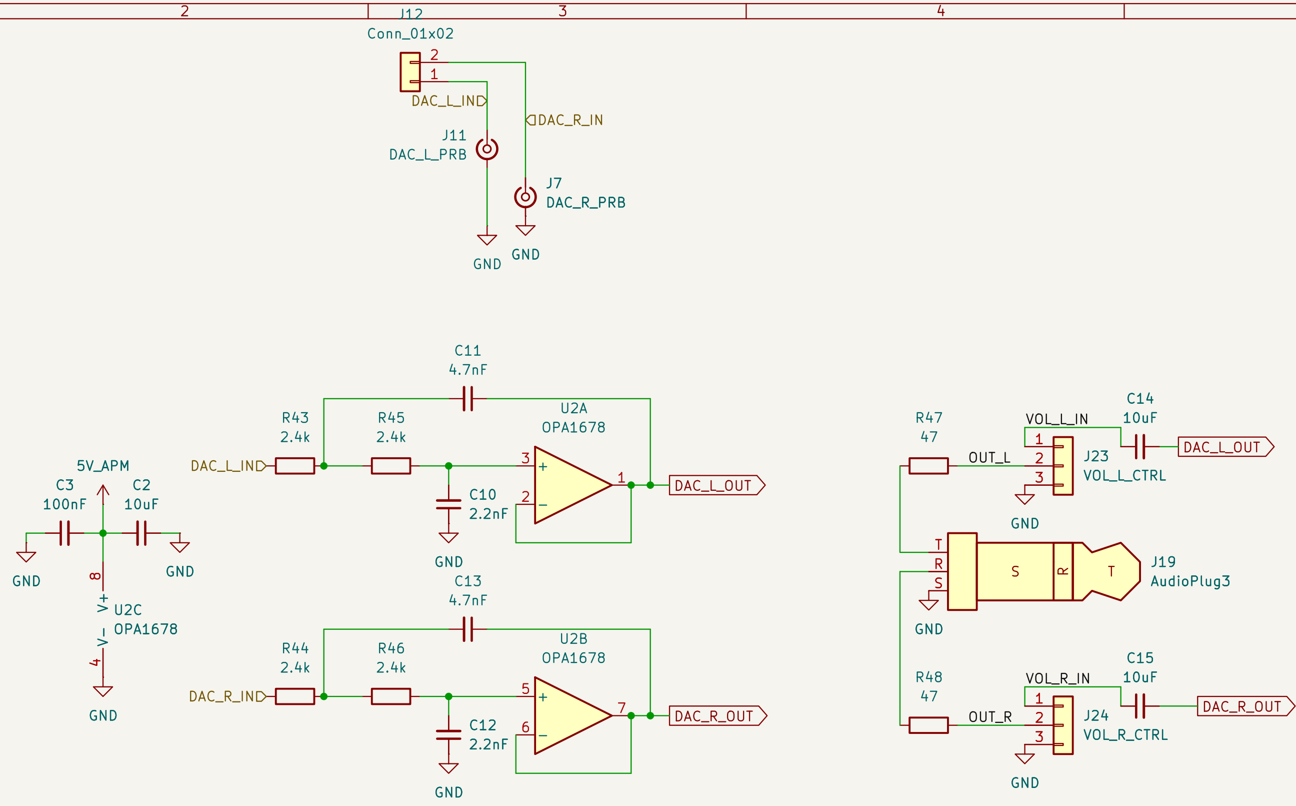


Sheet: /THERMISTORS_APM/
File: THERMISTORS_APM.kicad_sch

Title:

Size: A4	Date:	**Rev:**
KiCad E.D.A. 10.0.4		Id: 5/8


C6
100nF
3V3_APM
GND
SPIFLASH_CSD
SPIFLASH_SCKD
SPIFLASH_MOSID
SPIFLASH_MISOD
SPIFLASH_END
3V3_APM
R14
10k
U4
W25Q128JVS
C8
100nF
3V3_APM
GND
QSPIFLASH_CSD
QSPIFLASH_CLKD
QSPIFLASH_IO0D
QSPIFLASH_IO1D
QSPIFLASH_IO2D
QSPIFLASH_IO3D
U6
W25Q128JVS
C4
100nF
3V3_APM
U5
MR25H40
GND
R6
10k
3V3_APM
SPIMRAM_END
SPIMRAM_CSD
SPIMRAM_SCK
SPIMRAM_MISO
SPIMRAM_MOSI
I2C addr: 0x50 & 0x51
U8
24LC1025-I_P
JP3
A0
JP4
A1
JP5
A2
3V3_APM
R8
10k
R17
10k
R18
10k
C16
100nF
EEPROM_SCLD
EEPROM_END
EEPROM_SDAD
Sheet: /STORAGE_APM/
File: STORAGE_APM.kicad_sch
Title:
Size: A4
Date:
Rev:
KiCad E.D.A. 10.0.4
Id: 6/8



Sheet: /AUDIO_APM/
File: AUDIO_APM.kicad_sch

Title:

Size: A4 | Date: | **Rev:**

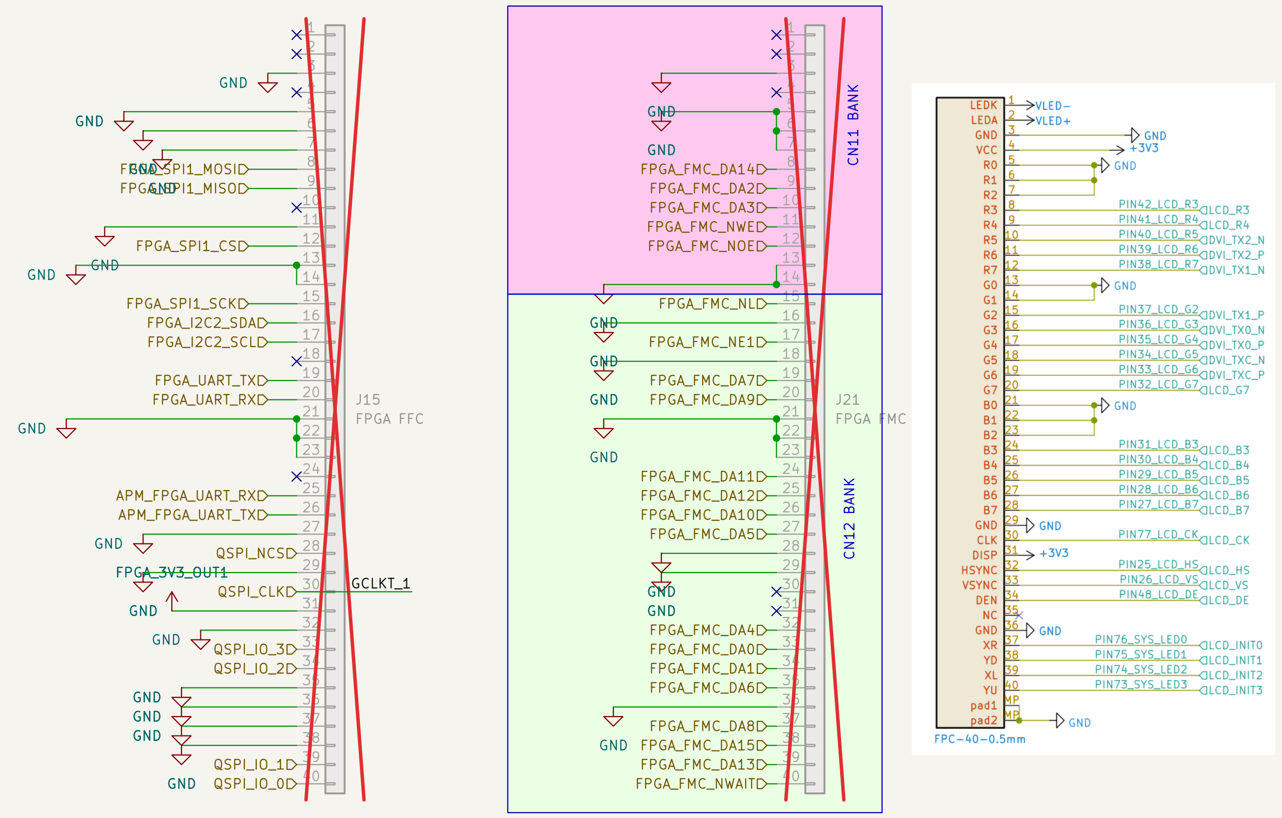
KiCad E.D.A. 10.0.4 | Id: 7/8

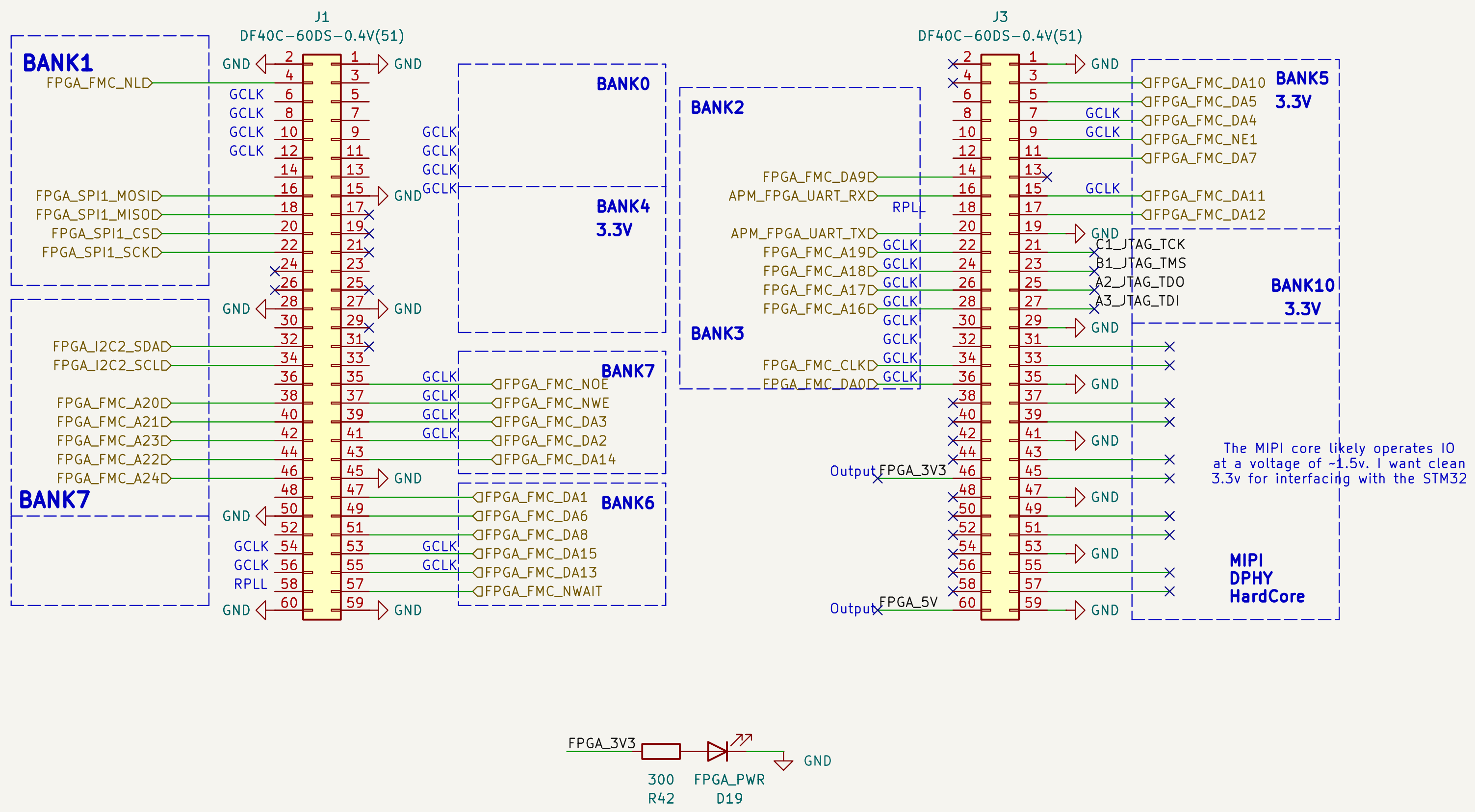
ACM FPGA Interface

Dual FPC STM32<–>Tang Nano 20K FPGA connection configuration. The STM32 side has two connectors where only one can be active at a time

- Configuration 1: FPGA_FFC with SPI, QSPI, UART connections.
- Configuration 2: FPGA_FMC with:
    - 16 bit muxed data
    - and async FMC

configuration resulting in a 64MB shared address space between the STM32 & FPGA





Sheet: /FPGA_INTERFACE/
File: FPGA_INTERFACE.kicad_sch

Title:

Size: A3	Date:	Rev:
KiCad E.D.A. 10.0.4		Id: 8/8